HACKATHON · CREATIVITY AND PROBLEM SOLVING

AI has no ideas.
You do.

Practical models for better ideas, for using AI without being replaced by it, and for designing with pen and paper.

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BEFORE WE START

Today every team has access to the same technology.

So AI can’t be your advantage.

The paradox of abundance

BEFORE

Having ideas took effort.

Whoever had more of them started ahead.

TODAY

A hundred ideas in ten seconds.

All plausible. All similar to everyone else's.

Creativity is your brain thanking you for having studied and thought enough.

Value shifts from the answer to the question.

01 · TRIZ

Inventions repeat a few patterns.

Starting in 1946, Genrich Altshuller analyzed tens of thousands of patents and found that brilliant solutions follow regularities that can be learned.

> A hard problem is a **contradiction**.
>
> A compromise softens it. An invention removes it.

“If I improve A, B gets worse”

Find the contradiction in your field: that’s where the idea is.

I improve…	…and it worsens	Compromise	Inventive solution
Security	Ease of access	simpler passwords	Passkeys: nothing to remember
Features	Simplicity	deeper menus	Interface that appears only when needed
Real time	Server load	refresh every 30 seconds	Push only when something changes
Personalization	Privacy	ask consent for everything	On-device model: data never leaves

6 of the 40 principles, translated into software

#1

Segmentation

Split into independent parts: modules, microservices, feature flags.

#10

Preliminary action

Do now what you'll need later: caching, precomputing, prefetching.

#13

The other way round

Do the opposite: pull instead of push, the user becomes the producer.

#24

Intermediary

Add a mediator: proxy, adapter, an AI agent between user and system.

#25

Self-service

The system serves itself: self-healing, user-generated content.

#27

Cheap copies

Use disposable copies: mocks, prototypes, digital twins.

Exercise: 5 minutes, one sheet

1. **The problem in one sentence:** who struggles, with what, when.
2. **Contradiction:** “If I improve ___, ___ gets worse.”
3. **Ideal result:** “The problem solves itself when…”
4. **Three principles, one solution idea each** (e.g. #1, #13, #25).
5. **Resources to exploit:** data, users, time, hardware you already have.

02 · AI AS A SPARRING PARTNER

Don’t ask for answers. Ask for resistance.

PASSIVE PROMPT

“Give me 10 ideas for a sustainability app.”

You get the internet’s average. The other teams are getting the same ones.

ACTIVE PROMPT

“Here are our 3 ideas and the contradiction we found. Attack the weakest: what are we taking for granted?”

You get an opponent that makes you think better.

Active prompts to copy for this challenge

The interviewer	Don’t suggest solutions. Ask me 7 questions, one at a time, to fully understand the problem [X]. After each answer, tell me what’s still unclear to you.
The TRIZ translator	Use the TRIZ method and its 40 principles. Our problem is [X]. Rephrase it as “improving A worsens B”, then apply the TRIZ principles you see fit, explain why they would work and give me one concrete idea for each.
The devil’s advocate	Here’s our idea: [X]. You’re a skeptical judge at this hackathon: give me 3 reasons it won’t work and the most fragile assumption we’re taking for granted.

Active prompts to escape the average

The forced analogy	How would an emergency room, an ant colony and a video game solve [X]? For each one, extract the mechanism, not the look.
The extreme constraint	How would you build [X] with a tenth of the code? With no screen? With only 2 hours of work?
The mirror	Here are our notes: [photo or text]. Don’t add ideas: tell me what patterns you see and what contradictions, errors or gaps you spot.

THE 10-MINUTE RULE

First 10 minutes of heads and paper.
Then AI.

If you ask AI before you think, its first answer becomes your fence.

03 · DESIGNING WITHOUT A SCREEN

Paper, markers, sticky notes. That’s it.

Crazy 8s

A sheet folded into 8, eight sketches in 8 minutes. Quantity before quality.

Brainwriting 6-3-5

3 ideas each in 5 minutes, then pass the sheet on and build on the others’ ideas.

6-panel storyboard

The user before, during and after. If it doesn’t fit in 6 panels, it won’t fit in a demo.

Paper prototype

Draw the screens, a teammate plays “the computer”. Test before writing any code.

HAPPY HACKING

AI multiplies.

But zero times anything is still zero.

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